POWER BI – WEEK 9 PROJECT REPORT

HOUSING DATA CLEANING AND TRANSFORMATION USING MICROSOFT POWER BI

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B.SC DS & AI

1. Introduction

Power BI is a Business Intelligence tool used to analyze and visualize data. This project focuses on cleaning and transforming housing sales data using Power BI and Power Query Editor. The main purpose is to prepare the dataset for further analysis.

2. Objectives

- To import housing sales data into Power BI.
- To clean and organize the dataset.
- To remove duplicate records.
- To filter unnecessary rows and columns.
- To change data types and rename columns.
- To create calculated columns for analysis.

3. Tools Used

- Microsoft Power BI Desktop
- Power Query Editor
- Housing Sales Dataset – kc_house_data

4. Dataset Description

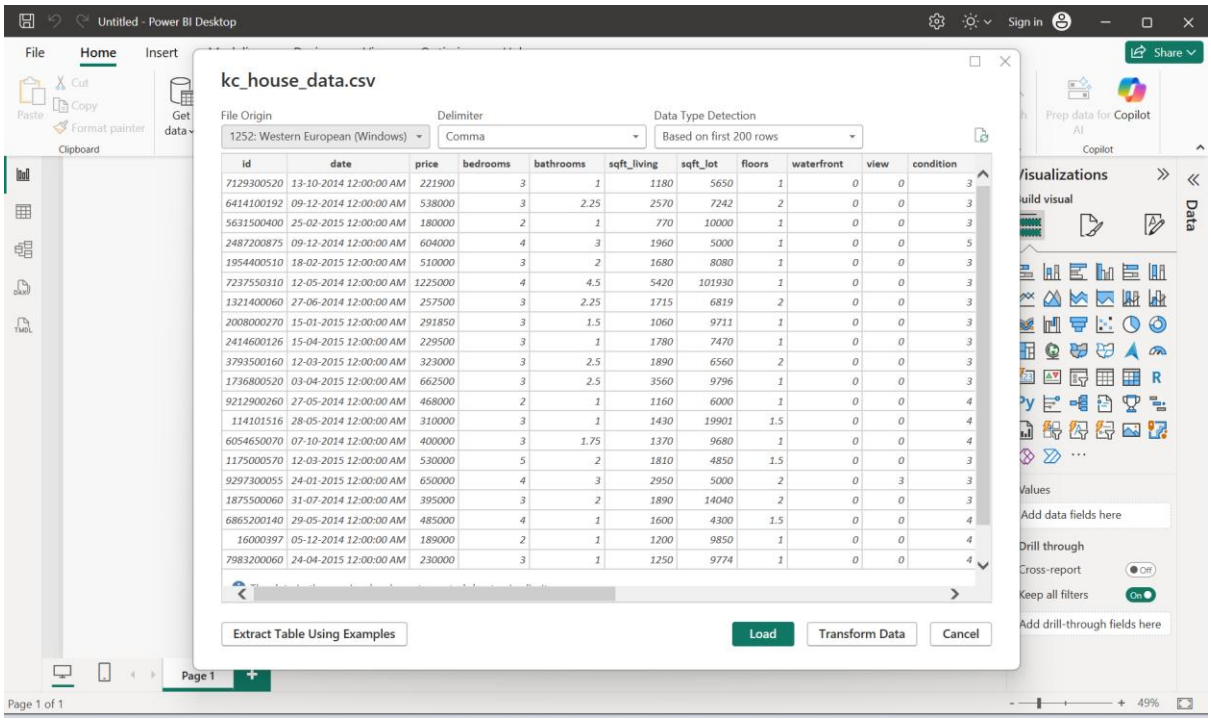
The project uses a housing sales dataset containing property details and sales information.

The dataset includes columns such as:

- ID
- Date
- Price
- Bedrooms
- Bathrooms
- Living Area
- Lot Area
- Year Built

5. Data Import

The housing sales dataset was imported into Microsoft Power BI. The dataset was opened in Power Query Editor to perform data cleaning and transformation.



6. Data Cleaning

6.1 Promoting Headers

The first row of the dataset was promoted as column headers to make the data easier to understand.

6.2 Changing Data Types

The data types of the columns were changed according to their values. Date, whole number, decimal number, and text data types were assigned where required.

6.3 Removing Duplicates

Duplicate records were removed to avoid repeated data and improve data quality.

6.4 Filtering Rows

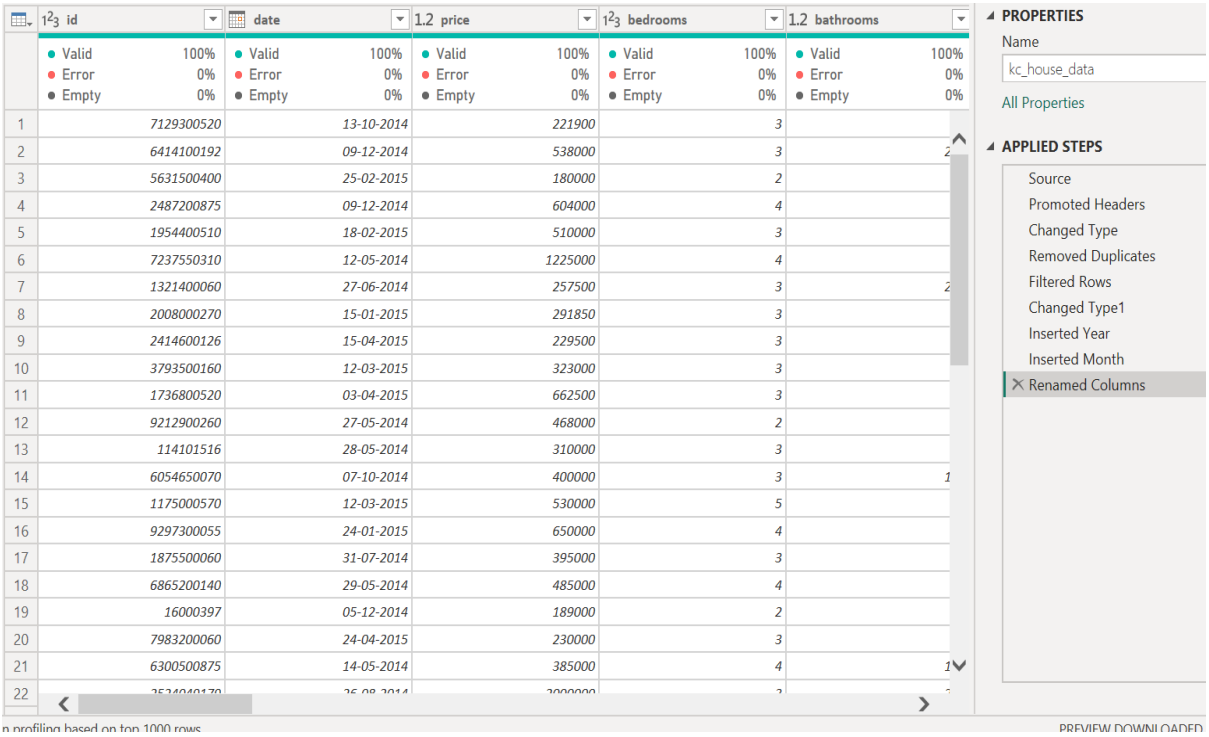
Unnecessary rows were filtered based on the project requirements.

6.5 Removing Unnecessary Columns

Columns that were not required for the analysis were removed from the dataset.

6.6 Renaming Columns

Selected column names were renamed to make the dataset more understandable.



	id	date	price	bedrooms	bathrooms
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	7129300520	13-10-2014	221900	3	
2	6414100192	09-12-2014	538000	3	2
3	5631500400	25-02-2015	180000	2	
4	2487200875	09-12-2014	604000	4	
5	1954400510	18-02-2015	510000	3	
6	7237550310	12-05-2014	1225000	4	
7	1321400060	27-06-2014	257500	3	2
8	2008000270	15-01-2015	291850	3	
9	2414600126	15-04-2015	229500	3	
10	3793500160	12-03-2015	323000	3	
11	1736800520	03-04-2015	662500	3	
12	9212900260	27-05-2014	468000	2	
13	114101516	28-05-2014	310000	3	
14	6054650070	07-10-2014	400000	3	1
15	1175000570	12-03-2015	530000	5	
16	9297300055	24-01-2015	650000	4	
17	1875500060	31-07-2014	395000	3	
18	6865200140	29-05-2014	485000	4	
19	16000397	05-12-2014	189000	2	
20	7983200060	24-04-2015	230000	3	
21	6300500875	14-05-2014	385000	4	1
22					

PROPERTIES
Name: kc_house_data
All Properties

APPLIED STEPS
Source
Promoted Headers
Changed Type
Removed Duplicates
Filtered Rows
Changed Type1
Inserted Year
Inserted Month
Renamed Columns

n profiling based on top 1000 rows
PREVIEW DOWNLOADED

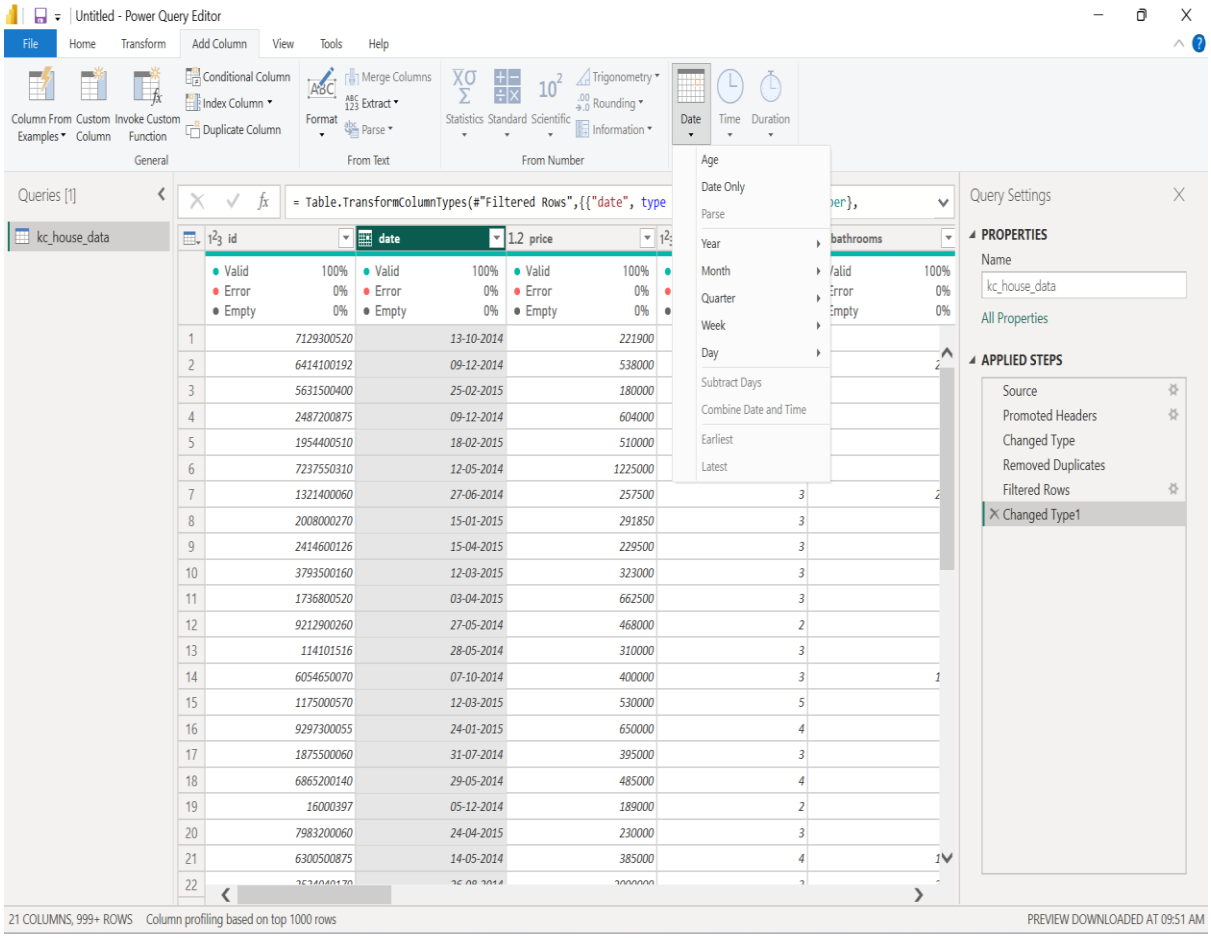
7. Creating Custom Columns

7.1 Sale Year

A new column named Sale Year was created by extracting the year from the sale date.

7.2 Sale Month

A new column named Sale Month was created by extracting the month from the sale date.

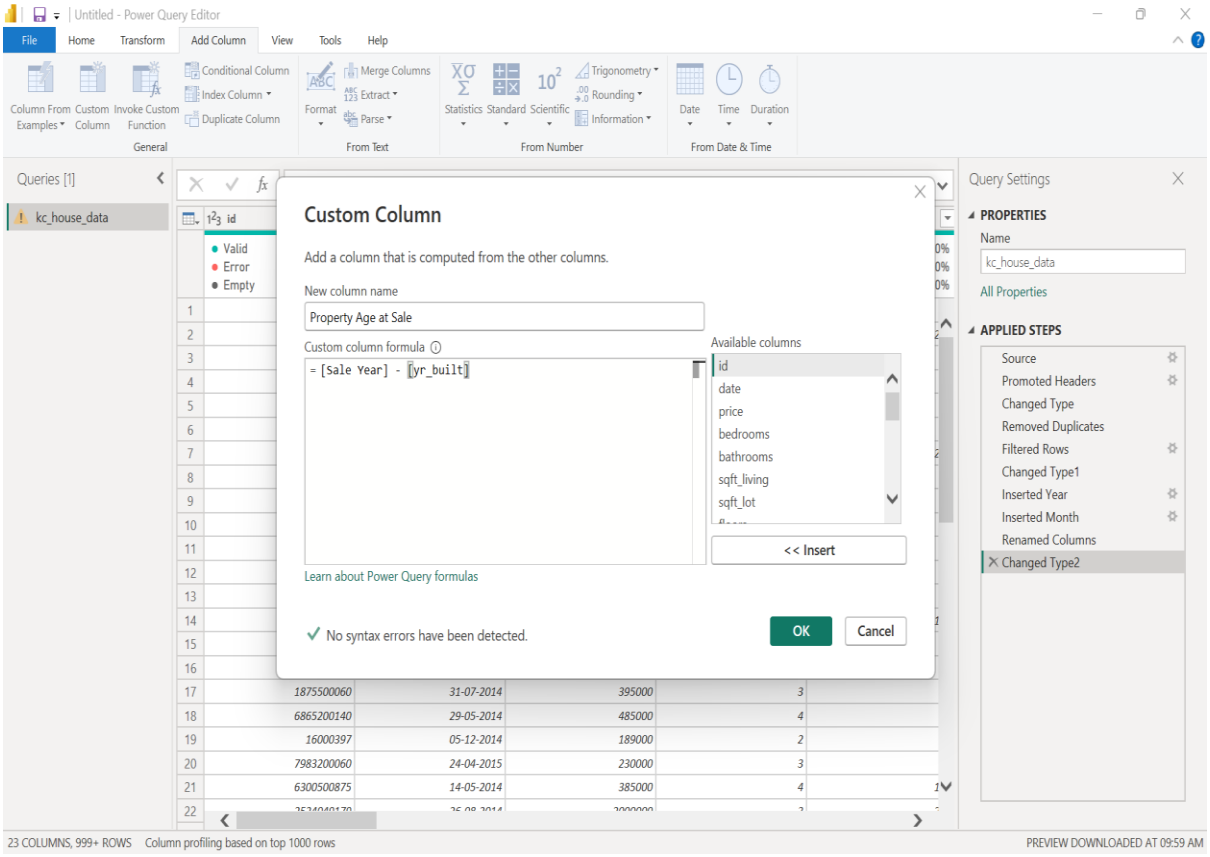


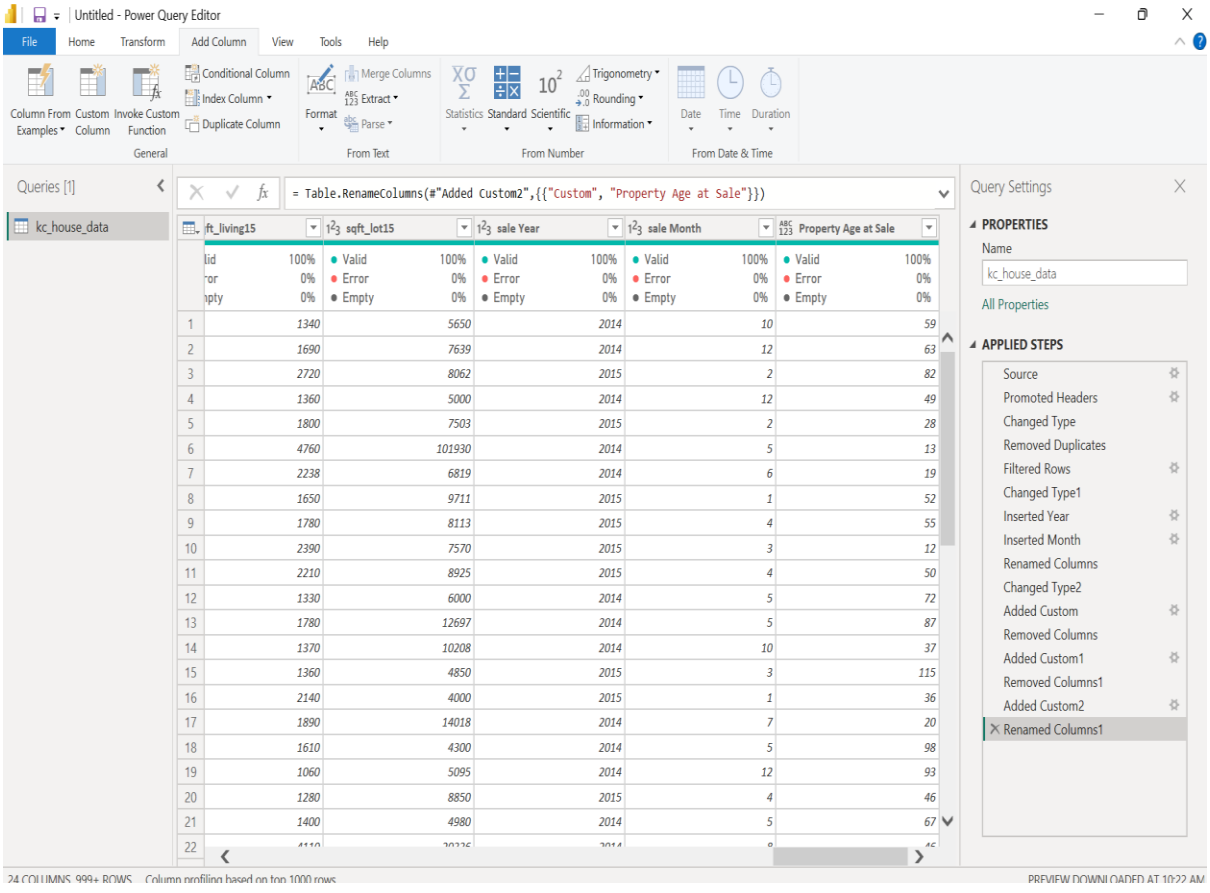
7.3 Property Age at Sale

A custom column named Property Age at Sale was created to calculate the age of the property at the time of sale.

Formula:

Property Age at Sale = Sale Year - Year Built


Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [1]
kc_house_data
Custom Column
Add a column that is computed from the other columns.
New column name
Property Age at Sale
Custom column formula
= [Sale Year] - [yr_built]
Available columns
id
date
price
bedrooms
bathrooms
sqft_living
sqft_lot
<< Insert
Learn about Power Query formulas
No syntax errors have been detected.
OK
Cancel
Query Settings
PROPERTIES
Name
kc_house_data
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Removed Duplicates
Filtered Rows
Changed Type1
Inserted Year
Inserted Month
Renamed Columns
Changed Type2
23 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 09:59 AM

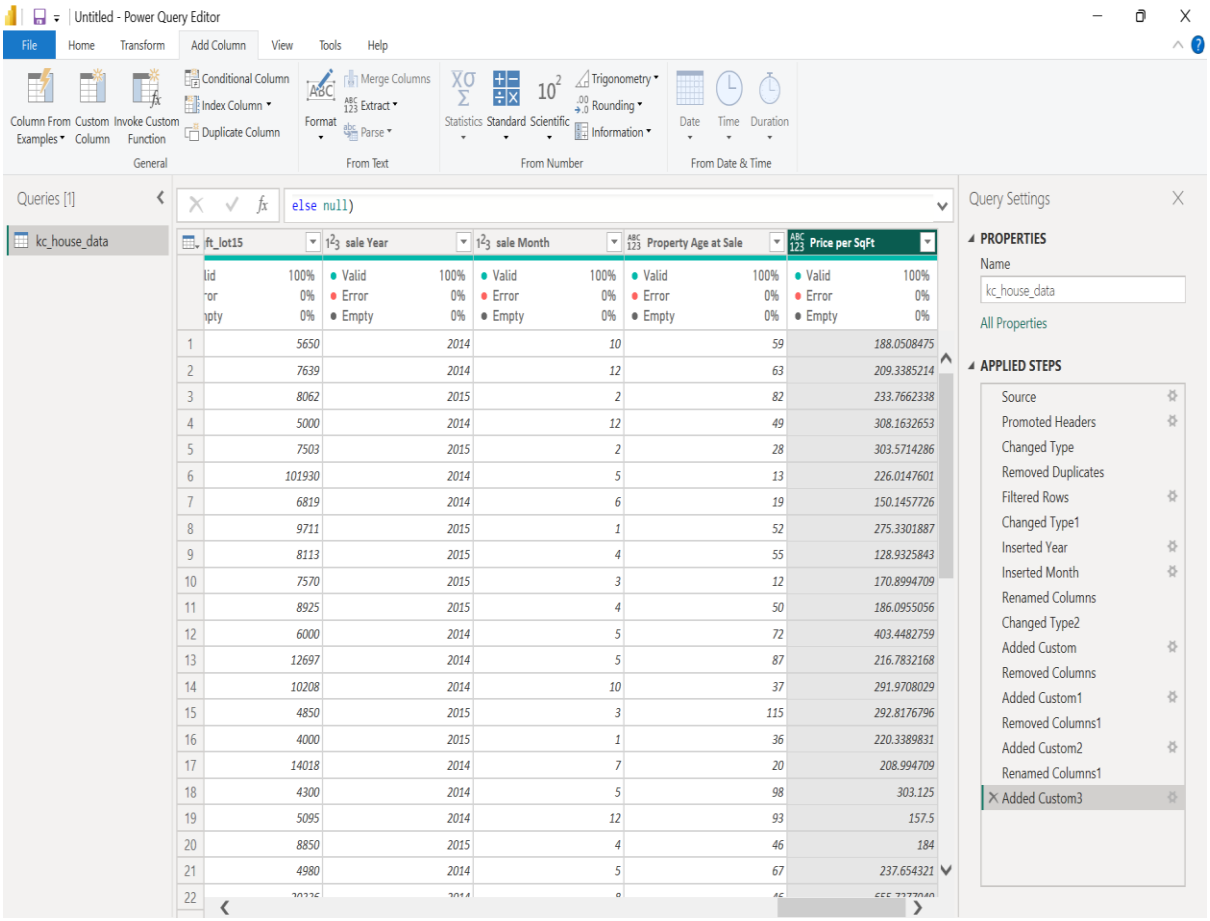

Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Queries [1]
kc_house_data
= Table.RenameColumns(#"Added Custom2",{{"Custom", "Property Age at Sale"}})
sqft_living15
sqft_lot15
sale Year
sale Month
Property Age at Sale
Query Settings
PROPERTIES
Name
kc_house_data
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Removed Duplicates
Filtered Rows
Changed Type1
Inserted Year
Inserted Month
Renamed Columns
Changed Type2
Added Custom
Removed Columns
Added Custom1
Removed Columns1
Added Custom2
Renamed Columns1
24 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 10:22 AM

7.4 Price per SqFt

A custom column named Price per SqFt was created to calculate the price per square foot.

Formula:

Price per SqFt = Price / SqFt Living

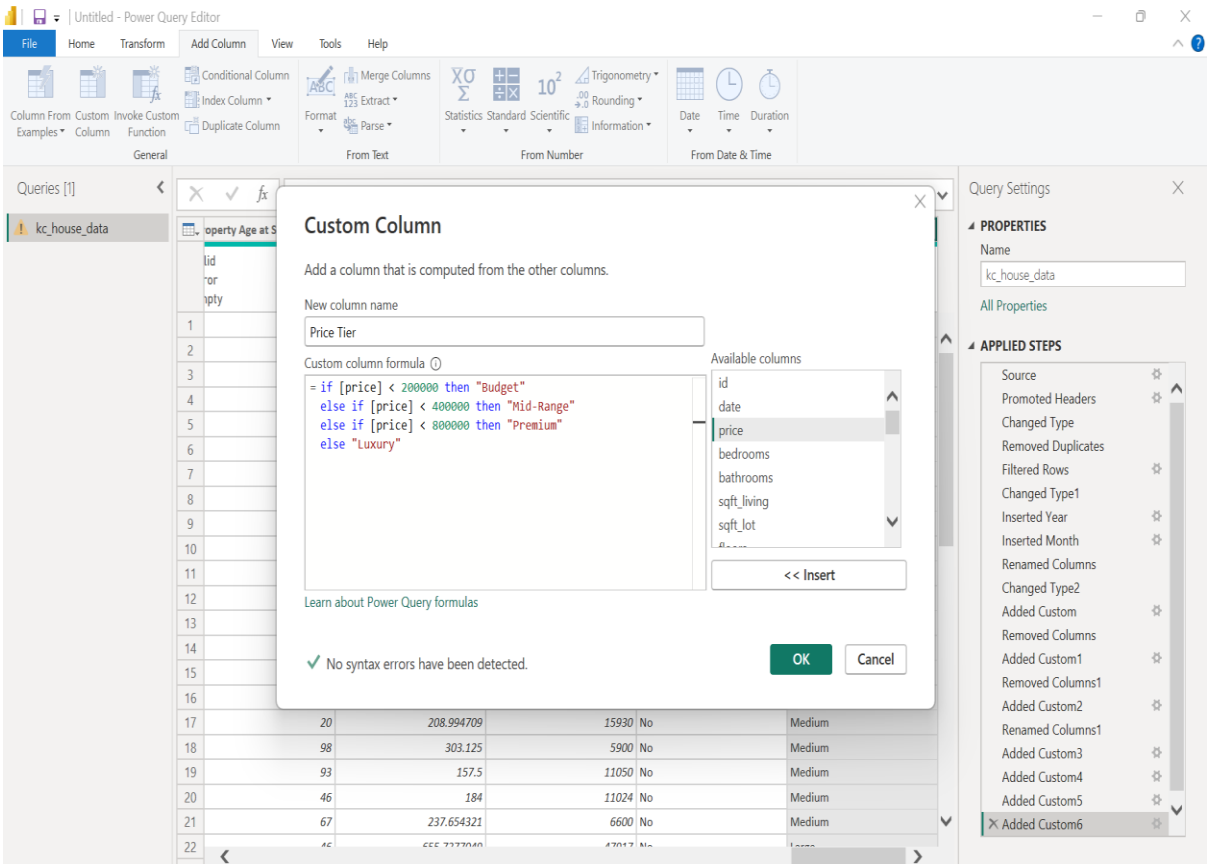


7.5 Price Tier

A custom column named Price Tier was created to classify houses based on their selling prices. Different price

categories were used to group properties according to their price ranges.

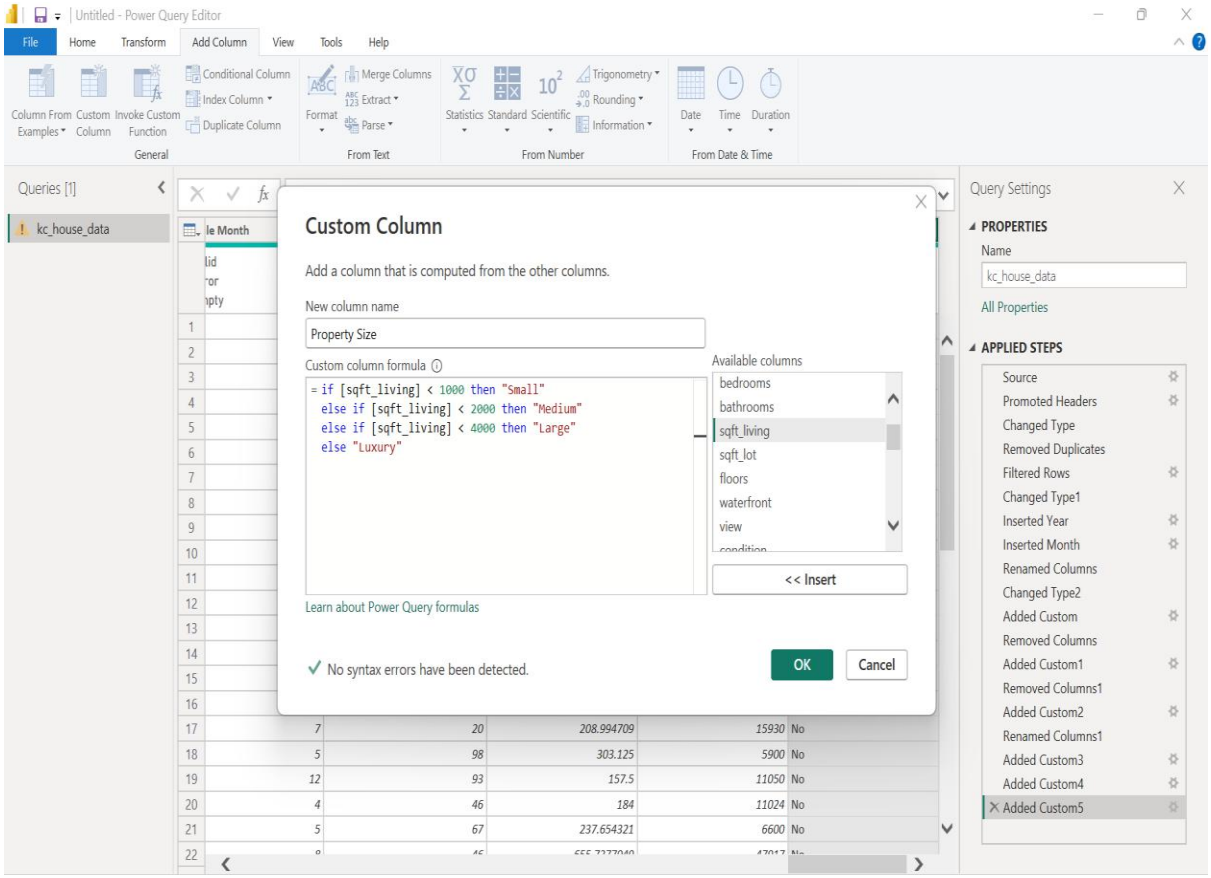
This classification helps identify lower-priced, medium-priced, and higher-priced properties. It also makes it easier to compare property prices and understand the distribution of houses in the dataset.



7.6 Property Size

A custom column named Property Size was created to categorize houses based on their living area. Properties were grouped into different size categories according to their square footage.

This classification helps identify smaller, medium-sized, and larger properties. It makes property comparison easier and provides a clear understanding of the size distribution in the housing dataset.

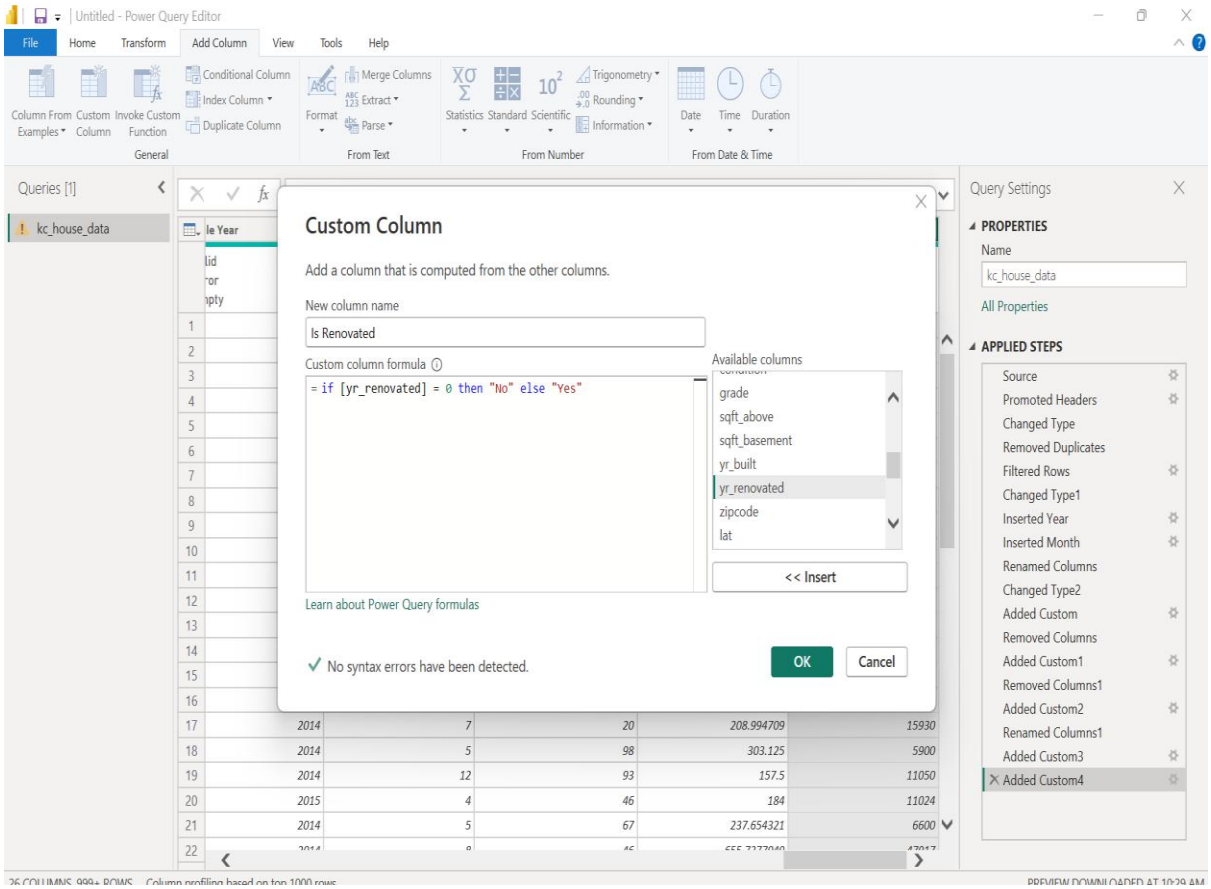


7.7 Is Renovated

A custom column named Is Renovated was used to identify whether a property had been renovated or not. The yr_renovated column was used to check the renovation details of each property.

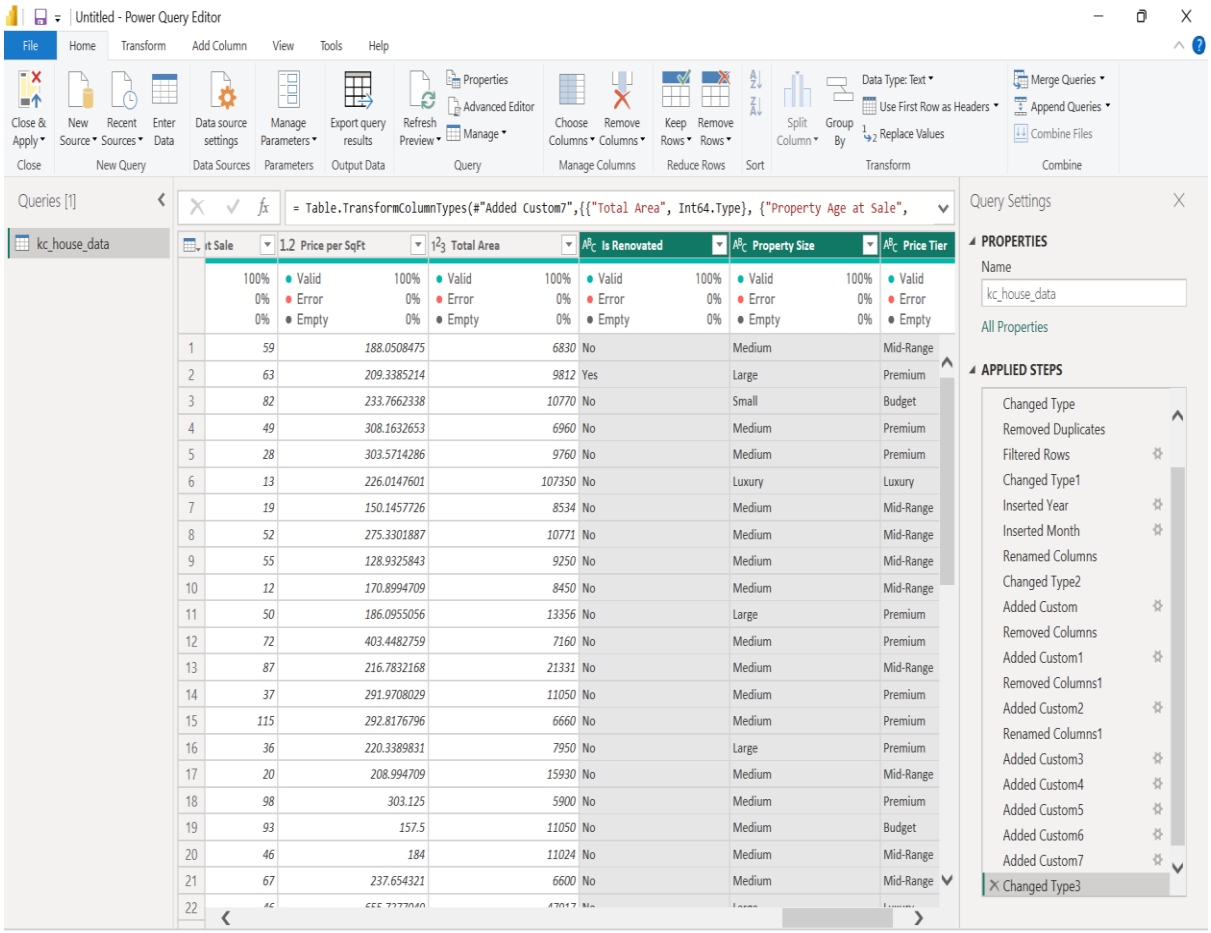
Properties with a renovation year were classified as renovated, while properties without a renovation year were classified as not renovated.

This column helps to compare renovated and non-renovated properties and understand the renovation status of houses in the dataset.



8. Applied Steps

The Applied Steps section in Power Query Editor displays the sequence of transformations performed on the dataset. It helps track and review the data cleaning process.



		1.2 Price per SqFt	Total Area	Is Renovated	Property Size	Price Tier
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	Valid Error Empty
1	59	188.0508475	6830	No	Medium	Mid-Range
2	63	209.3385214	9812	Yes	Large	Premium
3	82	233.7662338	10770	No	Small	Budget
4	49	308.1632653	6960	No	Medium	Premium
5	28	303.5714286	9760	No	Medium	Premium
6	13	226.0147601	107350	No	Luxury	Luxury
7	19	150.1457726	8534	No	Medium	Mid-Range
8	52	275.3301887	10771	No	Medium	Mid-Range
9	55	128.9325843	9250	No	Medium	Mid-Range
10	12	170.8994709	8450	No	Medium	Mid-Range
11	50	186.0955056	13356	No	Large	Premium
12	72	403.4482759	7160	No	Medium	Premium
13	87	216.7832168	21331	No	Medium	Mid-Range
14	37	291.9708029	11050	No	Medium	Premium
15	115	292.8176796	6660	No	Medium	Premium
16	36	220.3389831	7950	No	Large	Premium
17	20	208.994709	15930	No	Medium	Mid-Range
18	98	303.125	5900	No	Medium	Premium
19	93	157.5	11050	No	Medium	Budget
20	46	184	11024	No	Medium	Mid-Range
21	67	237.654321	6600	No	Medium	Mid-Range
22						

9. Final Output

After completing all the data cleaning and transformation operations, the housing dataset became more organized and structured. Unnecessary records and columns were removed, appropriate data types were assigned, and new columns were created to provide additional information about property sales.

The final transformed dataset is ready for further analysis and visualization using Power BI.

10. Conclusion

This project provided practical knowledge of data cleaning and transformation using Microsoft Power BI. Through this project, I learned how to import datasets, promote headers, change data types, remove duplicates, filter rows, remove unnecessary columns, and rename columns.

I also learned how to create custom columns such as Sale Year, Sale Month, Property Age at Sale, Price per SqFt, Price Tier, and Property Size. Overall, this project improved my understanding of data preparation and the importance of organizing raw data before performing analysis and visualization.